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Editors
Transportation Research Part E: Logistics and Transportation Review

Dear Editors,

Please consider the enclosed manuscript, "Cost, Carbon, and the Speed of Replenishment in Distribution Network Design," for publication in *Transportation Research Part E: Logistics and Transportation Review*.

Firms continue to revisit how many distribution centers to operate and how fast to replenish them. The reasons they give are mainly cost and delivery service; carbon pricing and emissions reporting now bear on such decisions as well. The manuscript models the two decisions jointly, under a carbon price or cap and a service requirement. Slower line haul burns less fuel but requires more safety stock and more stock in transit; more distribution centers shorten the last mile but add facilities and give up the pooling that keeps inventory low.

We show that the design problem has a unique optimum at every carbon price. For a given network, carbon favors faster replenishment when the inventory that speed saves emits more per dollar than the speed itself; because the capital tied up in goods in transit carries no carbon, financing those goods works toward slower replenishment. For a given speed, carbon favors more distribution centers when last-mile delivery emits more per dollar than facilities and lost pooling. In the joint design, the direction of density, and of speed when the two comparisons agree, follows from four carbon intensities per dollar and two shares that the cost-minimizing design reveals, which a firm can compute from its own rate data; a small carbon price can move density the other way first. A calibration to public US data shows how a carbon price moves a reference network: at prices now in use it mainly slows the line haul, and the number of distribution centers rises only slightly.

The manuscript fits the journal's focus on logistics and operations models with application and on the evaluation of public policies. It is original, has not been published, and is not under consideration elsewhere.

Sincerely,

Ram Ganeshan